

5.5 For each of the following two time-series regression models, and assuming MR1-MR6 hold, find $\text{var}(b_2|x)$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|x) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

- c. Provide an intuitive explanation for these results.

$$y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2} + \dots + \beta_k x_{ik} + e_i, e_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

Assumption: (1) Strict Exogeneity: $E(e_i|x) = 0$ (2) Conditional Homoskedasticity: $\text{Var}(e_i|x) = \sigma^2$ (3) Conditional Uncorrelated Errors: $\text{Cov}(e_i, e_j|x) = 0 \quad i \neq j$

(4) No exact linear relationship exists between the explanatory variables (5) $e_i|x \sim N(0, \sigma^2)$

$$\sum (t_i - \bar{t}) = \sum t_i - n\bar{t} = n\bar{t} - n\bar{t} = 0 \dots (*)$$

$$\text{Var}(y_i|x) = \text{Var}(\beta_1 + \beta_2 t + e_i|x) = \text{Var}(e_i) = \sigma^2$$

$$a. \text{var}(b_2|x) = \text{var}\left(\frac{\sum_{i=1}^T (t_i - \bar{t})(y_i - \bar{y})}{\sum_{i=1}^T (t_i - \bar{t})^2}\right) = \frac{1}{\left(\sum_{i=1}^T (t_i - \bar{t})^2\right)^2} \text{var}\left(\sum_{i=1}^T (t_i - \bar{t}) y_i - \bar{y} \sum_{i=1}^T (t_i - \bar{t})\right) \stackrel{(*)}{=} \frac{1}{\left[\sum_{i=1}^T (t_i - \bar{t})^2\right]^2} \text{var}\left(\sum_{i=1}^T (t_i - \bar{t}) y_i\right)$$

$$\stackrel{MR4}{=} \frac{1}{\left(\sum_{i=1}^T (t_i - \bar{t})^2\right)^2} \sum_{i=1}^T \text{var}((t_i - \bar{t})(\beta_1 + \beta_2 t_i + e_i)) = \frac{1}{\left(\sum_{i=1}^T (t_i - \bar{t})^2\right)^2} \sum_{i=1}^T (t_i - \bar{t})^2 \text{var}(y_i) = \frac{\sigma^2 \sum_{i=1}^T (t_i - \bar{t})^2}{\left(\sum_{i=1}^T (t_i - \bar{t})^2\right)^2} = \frac{\sigma^2}{\sum_{i=1}^T (t_i - \bar{t})^2}$$

$$= \sigma^2 \left(\frac{1}{\sum_{i=1}^T t^2} - \frac{1}{T \left(\sum_{i=1}^T t\right)^2} \right) = \sigma^2 \left(\frac{1}{\frac{T(T+1)(2T+1)}{6}} - \frac{1}{T \cdot \frac{T^2(T+1)^2}{4}} \right) = \sigma^2 \left[\frac{12}{T(T+1)(2T+1)} - \frac{4}{T^2(T+1)^2} \right] = \frac{12\sigma^2}{T(T+1)} \cdot \frac{1}{4T+2-3T-3} = \frac{12\sigma^2}{T(T+1)(T-1)}$$

$$\lim_{T \rightarrow \infty} \text{var}(b_2|x) = \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T(T+1)(T-1)} = \frac{12\sigma^2}{\infty} = 0 \Rightarrow \text{least square estimator } (b_2) \text{ is consistent}$$

$$b. \sum_{i=1}^T (0.5)^{2t} = \sum_{i=1}^T \left(\frac{1}{4}\right)^t = \frac{\frac{1}{4} \left(1 - \left(\frac{1}{4}\right)^T\right)}{1 - \frac{1}{4}} = \frac{1}{3} \left[1 - 4^{-T}\right] \quad \sum_{i=1}^T 0.5^{2t} = \frac{\frac{1}{2} \left(1 - \left(\frac{1}{2}\right)^T\right)}{1 - \frac{1}{2}} = 1 - 2^{-T}$$

$$\text{var}(b_2|x) = \sigma^2 \left[\frac{1}{\sum_{i=1}^T ((0.5)^t - \bar{t})^2} \right] = \sigma^2 \left[\frac{1}{\sum_{i=1}^T (0.5)^{2t} - \frac{1}{T} \left(\sum_{i=1}^T (0.5)^t\right)^2} \right] = \sigma^2 \left[\frac{1}{\frac{1}{3} [1 - 4^{-T}] - \frac{1}{T} [1 - 2^{-T}]^2} \right] = \sigma^2 \cdot \frac{3T}{T[1 - (\frac{1}{4})^T] - 3[1 - (\frac{1}{2})^T]^2}$$

$$= \sigma^2 \left[\frac{3T}{T[1 - (\frac{1}{4})^T] - 3[1 - (\frac{1}{2})^T + (\frac{1}{2})^{2T}]} \right]$$

$$\lim_{T \rightarrow \infty} \text{var}(b_2|x) = \lim_{T \rightarrow \infty} \sigma^2 \left[\frac{3T}{T[1 - (\frac{1}{4})^T] - 3[1 - (\frac{1}{2})^T + (\frac{1}{2})^{2T}]} \right] = \lim_{T \rightarrow \infty} \frac{3\sigma^2}{[1 - (\frac{1}{4})^T] - \frac{3}{T} [1 - (\frac{1}{2})^T + (\frac{1}{2})^{2T}]} = \frac{3\sigma^2}{[1 - 0] - 0 \cdot [1 - 0 + 0]} = 3\sigma^2$$

$\Rightarrow$ least square estimator is NOT consistent

c (a) As T increases, we have more information to look for coefficients, hence $\lim_{T \rightarrow \infty} \text{var}(b_2|x) = 0$ and can estimate β_2 precisely.

(b) As T increases, $(0.5)^T \rightarrow 0$, the latter doesn't have more information to look for coefficient

, hence $\lim_{T \rightarrow \infty} \text{var}(b_2|x) \neq 0$ and not consistent.

- 5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- $\beta_2 = 0$
- $\beta_1 + 2\beta_2 = 5$
- $\beta_1 - \beta_2 + \beta_3 = 4$

Note that

$$\widehat{\text{Cov}}(b_1, b_2, b_3) = \begin{pmatrix} \widehat{\text{var}}(b_1) & \widehat{\text{cov}}(b_1, b_2) & \widehat{\text{cov}}(b_1, b_3) \\ \widehat{\text{cov}}(b_2, b_1) & \widehat{\text{var}}(b_2) & \widehat{\text{cov}}(b_2, b_3) \\ \widehat{\text{cov}}(b_3, b_1) & \widehat{\text{cov}}(b_3, b_2) & \widehat{\text{var}}(b_3) \end{pmatrix}$$

a. $H_0: \beta_2 = 0$ v.s. $H_1: \beta_2 \neq 0$, $\alpha = 0.05$

Test Statistic: $\frac{b_2 - 0}{\text{se}(b_2)} = \frac{3}{\sqrt{\widehat{\text{var}}(b_2)}} = \frac{3}{\sqrt{4}} = 1.5$ Rejection region = $\{t: |t| > t_{0.025}(60) = 2.00298\}$

$\Rightarrow$ Test Statistic is NOT in Rejection, we don't have enough evidence to reject H_0 , we can't conclude that $\beta_2 \neq 0$

b. $H_0: \beta_1 + 2\beta_2 = 5$ v.s. $H_1: \beta_1 + 2\beta_2 \neq 5$, $\alpha = 0.05$ Rejection region = $\{t: |t| > t_{0.025}(60) = 2.00298\}$

$\text{se}(b_1 + 2b_2) = \sqrt{\widehat{\text{var}}(b_1 + 2b_2)} = \sqrt{\widehat{\text{var}}(b_1) + 4\widehat{\text{var}}(b_2) + 4\widehat{\text{cov}}(b_1, b_2)} = \sqrt{3 + 4 \cdot 4 + 4(-2)} = \sqrt{11}$

Test Statistic: $\frac{b_1 + 2b_2 - 5}{\text{se}(b_1 + 2b_2)} = \frac{2 + 2 \cdot 3 - 5}{\sqrt{11}} = \frac{3}{\sqrt{11}} = 0.9045$ & Rejection Region

$\Rightarrow$ Test Statistic is NOT in Rejection, we don't have enough evidence to reject H_0 , we can't conclude that $\beta_1 + 2\beta_2 \neq 5$

c. $H_0: \beta_1 - \beta_2 + \beta_3 = 4$ v.s. $H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$, $\alpha = 0.05$ Rejection region = $\{t: |t| > t_{0.025}(60) = 2.00298\}$

$\text{se}(b_1 - b_2 + b_3) = \sqrt{\widehat{\text{var}}(b_1 - b_2 + b_3)} = \sqrt{\widehat{\text{var}}(b_1) + \widehat{\text{var}}(b_2) + \widehat{\text{var}}(b_3) - 2\widehat{\text{cov}}(b_1, b_2) + 2\widehat{\text{cov}}(b_1, b_3) - 2\widehat{\text{cov}}(b_2, b_3)}$
 $= \sqrt{3 + 4 + 3 - 2(-2) + 2(1) - 2(0)} = \sqrt{16} = 4$

Test Statistic: $\frac{b_1 - b_2 + b_3 - 4}{\text{se}(b_1 - b_2 + b_3)} = \frac{2 - 3 - 1 - 4}{4} = -1.5$ & Rejection Region

$\Rightarrow$ Test Statistic is NOT in Rejection, we don't have enough evidence to reject H_0 , we can't conclude that $\beta_1 - \beta_2 + \beta_3 \neq 4$

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i$, $i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2, \text{mean}} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1-MR6 hold:

- Show that $\hat{\beta}_{2, \text{mean}}$ is unbiased.
- Derive an expression for $\text{var}(\hat{\beta}_{2, \text{mean}} | x)$.
- Write down an expression for $\text{var}(\hat{\beta}_{2, \text{mean}})$.
- Under what conditions will $\hat{\beta}_{2, \text{mean}}$ be a consistent estimator for β_2 ?

$$\bar{x}_1 = \frac{2}{N} \sum_{i=1}^{\frac{N}{2}} x_i \quad \bar{x}_2 = \frac{2}{N} \sum_{i=\frac{N}{2}+1}^N x_i \quad \bar{y}_1 = \frac{2}{N} \sum_{i=1}^{\frac{N}{2}} y_i \quad \bar{y}_2 = \frac{2}{N} \sum_{i=\frac{N}{2}+1}^N y_i$$

a. Note that $E(\bar{y}_j) = E\left(\frac{2}{N} \sum_{i=1}^{\frac{N}{2}} \beta_1 + \beta_2 x_i + e_i\right) = \frac{2}{N} E\left(\frac{N}{2} \beta_1 + \beta_2 \sum_{i=1}^{\frac{N}{2}} x_i + \sum_{i=1}^{\frac{N}{2}} e_i\right) \stackrel{\text{MR}}{=} \beta_1 + \beta_2 \frac{2}{N} \sum_{i=1}^{\frac{N}{2}} x_i + 0 = \beta_1 + \beta_2 \frac{\sum x_i}{N} = \beta_1 + \beta_2 \bar{x}_j$ for $j=1, 2$

$E(\hat{\beta}_{2, \text{mean}} | x) = E\left(\frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} | x\right) = \frac{1}{\bar{x}_2 - \bar{x}_1} E(\beta_1 + \beta_2 \bar{x}_2 - \beta_1 - \beta_2 \bar{x}_1) = \frac{\beta_2 (\bar{x}_2 - \bar{x}_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 \Rightarrow E(\hat{\beta}_{2, \text{mean}}) = E(E(\hat{\beta}_{2, \text{mean}} | x)) = E(\beta_2) = \beta_2$

$\Rightarrow \hat{\beta}_{2, \text{mean}}$ is an unbiased estimator

b. $\text{Var}(\bar{y}_1 | x) = \text{Var}\left(\frac{2}{N} \sum_{i=1}^{\frac{N}{2}} y_i\right) = \frac{4}{N^2} \text{Var}\left(\sum_{i=1}^{\frac{N}{2}} \beta_1 + \beta_2 x_i + e_i\right) = \frac{4}{N^2} \text{Var}\left(\sum_{i=1}^{\frac{N}{2}} e_i\right) \stackrel{\text{MR3.4}}{=} \frac{4}{N^2} \cdot \frac{N}{2} \sigma^2 = \frac{2\sigma^2}{N}$. Similarly, $\bar{y}_2 = \frac{2\sigma^2}{N}$

$\text{Cov}(\bar{y}_1, \bar{y}_2 | x) = \text{Cov}\left(\sum_{i=1}^{\frac{N}{2}} \beta_1 + \beta_2 x_i + e_i, \sum_{j=\frac{N}{2}+1}^N \beta_1 + \beta_2 x_j + e_j\right) = \text{Cov}\left(\sum_{i=1}^{\frac{N}{2}} e_i, \sum_{j=\frac{N}{2}+1}^N e_j\right) = 0$

$\therefore \text{Var}(\hat{\beta}_{2, \text{mean}} | x) = \text{Var}\left(\frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} | x\right) = \frac{1}{(\bar{x}_2 - \bar{x}_1)^2} \text{Var}(\bar{y}_2 - \bar{y}_1 | x) = \frac{\frac{2}{N} \sigma^2 + \frac{2}{N} \sigma^2}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$

c. Note that $E(E(Y|x)) = E(Y)$ and

$$\text{Var}(Y) = E(Y^2) - (E(Y))^2 = E(E(Y^2|x)) - (E(E(Y|x)))^2 = E(\text{Var}(Y|x) + E((E(Y|x))^2)) - (E(E(Y|x)))^2 = E(\text{Var}(Y|x)) + \text{Var}(E(Y|x))$$

$\Rightarrow \text{Var}(\hat{\beta}_{2, \text{mean}}) = E(\text{Var}(\hat{\beta}_{2, \text{mean}} | x)) + \text{Var}(E(\hat{\beta}_{2, \text{mean}} | x)) = E\left(\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right) + \text{Var}(\beta_2) = \frac{4\sigma^2}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] + 0 = \frac{4\sigma^2}{N} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right]$

d. $\lim_{N \rightarrow \infty} \text{Var}(\hat{\beta}_{2, \text{mean}} | x) = \lim_{N \rightarrow \infty} \frac{1}{N} \frac{4\sigma^2}{(\bar{x}_2 - \bar{x}_1)^2}$. Since σ^2 is constant, to let $\lim_{N \rightarrow \infty} \text{Var}(\hat{\beta}_{2, \text{mean}} | x) = 0$, $(\bar{x}_2 - \bar{x}_1) \rightarrow 0 \Rightarrow \sum_{j=\frac{N}{2}+1}^N x_j - \sum_{i=1}^{\frac{N}{2}} x_i \rightarrow 0$

Condition: Sum of each halves are not close enough, then $\lim_{N \rightarrow \infty} \text{Var}(\hat{\beta}_{2, \text{mean}} | x) = 0$ and $\hat{\beta}_{2, \text{mean}}$ is consistent

- e. Randomly generate observations on x from a uniform distribution on the interval $(0,10)$ for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:
- $\text{var}(b_2|x)$ and $\text{var}(\hat{\beta}_{2,\text{mean}}|x)$ where b_2 is the OLS estimator.
 - Estimates for $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ where s_x^2 is the sample standard deviation for x using N as a divisor.
- f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,\text{mean}}$ is consistent?
- g. Show that $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b-a)^2/12$.]
- h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,\text{mean}}$ be consistent? Why or why not?

e. $\text{var}(b_2|x) = \frac{\sigma^2}{\sum_{i=1}^N (x_i - \bar{x})^2}$ $\text{var}(\hat{\beta}_{2,\text{mean}}|x) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$

By set.seed(456), we have the following result

f. (i) As N become larger, $\text{var}(b_2|x)$, $\text{var}(\hat{\beta}_{2,\text{mean}}|x)$ become smaller and it means that $\hat{\beta}_{2,\text{mean}}$ is consistent

We can note that $\text{var}(b_2|x) < \text{var}(\hat{\beta}_{2,\text{mean}}|x)$ for $N = 100, 500, 1000, 5000$ it means that b_2 is more efficient

(ii) As N become larger, $E[\frac{1}{s_x^2}]$ and $E[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}]$ are getting close to 0.12 and 0.16, respectively

g. 1. Since $x \sim U(0,10)$, $\text{Var}(x) = \frac{(10-0)^2}{12} = 8.33 = E[\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}] = E(s_x^2) \dots (*)$
 $\because x_i \sim U(0,1)$ By weak law of large number $E(\frac{\sum_{i=1}^N x_i}{N}) \xrightarrow{p} E(x)$ and $E(\frac{\sum_{i=1}^N x_i^2}{N}) \xrightarrow{p} E(x^2)$
 Since $f(x) = x - x^2$ and x^2 are continuous
 By continuous mapping theorem, $E(s_x^2) = E(\frac{\sum_{i=1}^N x_i^2}{N}) - E(\bar{x})^2 \xrightarrow{p} E(x^2) - (E(x))^2 = \text{Var}(x)$
 Since $\frac{1}{x}$ is continuous at $x \neq 0$. Therefore, $(s_x^2)^{-1} \xrightarrow{p} (\text{Var}(x))^{-1} \stackrel{(*)}{=} \frac{1}{8.33} = 0.12$ and $E[(s_x^2)^{-1}] \xrightarrow{p} (\text{Var}(x))^{-1} \stackrel{(*)}{=} \frac{1}{8.33} = 0.12$

2. By order statistic, for k^{th} order statistic, $X_{(k)} \xrightarrow{p} F^{-1}(\frac{k}{N})$. $F^{-1}(p) = p$ quantile $\because x \sim U(0,10) \therefore F(x) = \frac{x}{10}$
 $\therefore X_{(5)} \xrightarrow{p} F^{-1}(0.5) = 10 \times 0.5 = 5$
 By weak law of large number, $\bar{x}_1 \xrightarrow{p} E[x | x \in (0,5)] = 2.5$, $\bar{x}_2 \xrightarrow{p} E[x | x \in (5,10)] = 7.5$ $\because x_1, x_2, \frac{1}{x}$ are continuous at $x_1, x_2 \in \mathbb{R}, x \neq 0$
 By continuous mapping theorem, $\bar{x}_2 - \bar{x}_1 \xrightarrow{p} 7.5 - 2.5 = 5$ and $\frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \xrightarrow{p} \frac{4}{25} = 0.16$ and $E[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}] \xrightarrow{p} \frac{4}{25} = 0.16$

h. $\because x_i \stackrel{iid}{\sim} U(0,10)$. By weak law of large number, $\bar{x}_1, \bar{x}_2 \xrightarrow{p} E(x)$ $\because x_1, x_2$ are continuous at $x_1, x_2 \in \mathbb{R}$
 By continuous mapping theorem $\bar{x}_2 - \bar{x}_1 \xrightarrow{p} 0$, from (d) $\lim_{N \rightarrow \infty} \text{var}(\hat{\beta}_{2,\text{mean}}|x) = \lim_{N \rightarrow \infty} \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} = \lim_{N \rightarrow \infty} \frac{4\sigma^2}{N \cdot 0} \neq 0$ (not exist)
 $\Rightarrow \hat{\beta}_{2,\text{mean}}$ is NOT consistent

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N = 100
(i) var(b2|x) ≈ 1.217169
    var(beta2_mean|x) ≈ 1.639935
(ii) E[(sx^2)^(-1)] ≈ 0.1217169
    E[4/(xbar2-xbar1)^2] ≈ 0.1639935

N = 500
(i) var(b2|x) ≈ 0.2409379
    var(beta2_mean|x) ≈ 0.3220183
(ii) E[(sx^2)^(-1)] ≈ 0.120469
    E[4/(xbar2-xbar1)^2] ≈ 0.1610091

N = 1000
(i) var(b2|x) ≈ 0.120161
    var(beta2_mean|x) ≈ 0.1604093
(ii) E[(sx^2)^(-1)] ≈ 0.120161
    E[4/(xbar2-xbar1)^2] ≈ 0.1604093

N = 5000
(i) var(b2|x) ≈ 0.02400637
    var(beta2_mean|x) ≈ 0.03201852
(ii) E[(sx^2)^(-1)] ≈ 0.1200318
    E[4/(xbar2-xbar1)^2] ≈ 0.1600926

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